

ROLL NO:250701573

EXPERIMENT NO:8 EXPERIMENT TO UNDERSTAND LINEAR REGRESSION FOR A
GIVEN DATA SET

PROGRAM:

```
import numpy as np
```

```
import pandas as pd
```

```
df=pd.read_csv('Salary_data.csv')
```

```
print(df)
```

```
df.info()
```

```
df.dropna(inplace=True)
```

```
df.info()
```

```
print(df.describe())
```

```
features=df.iloc[:,[0]].values
```

```
label=df.iloc[:,[1]].values
```

```
from sklearn.model_selection import train_test_split
```

```
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_state=0)
```

```
from sklearn.linear_model import LinearRegression
```

```
model=LinearRegression()
```

```

model.fit(x_train,y_train)

print(model.score(x_train,y_train))
print(model.score(x_test,y_test))
print(model.coef_)
print(model.intercept_)

import pickle
pickle.dump(model,open('SalaryPred.model','wb'))

model=pickle.load(open('SalaryPred.model','rb'))

yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)

print("Estimated Salary for {} years of experience is {}: ".format(yr_of_exp,Salary))

```

OUTPUT:

	YearsExperience	Salary
0	1.1	39200
1	1.3	42100
2	1.5	44800
3	2.0	47200
4	2.2	50100
5	2.9	54800
6	3.0	56600
7	3.2	59100
8	3.7	62400
9	3.9	64600
10	4.0	67100
11	4.1	68900
12	4.5	72400

13	4.7	74800
14	5.0	78100
15	5.2	80600
16	5.3	83500
17	5.9	87200
18	6.0	89900
19	6.8	95100
20	7.1	97800
21	7.9	103500
22	8.2	107200
23	8.7	112400
24	9.0	116000
25	9.5	120500
26	9.6	123800
27	10.0	127500
28	10.3	131200
29	10.5	135000

<class 'pandas.DataFrame'>

RangeIndex: 30 entries, 0 to 29

Data columns (total 2 columns):

#	Column	Non-Null Count	Dtype
0	YearsExperience	30 non-null	float64
1	Salary	30 non-null	int64

dtypes: float64(1), int64(1)

memory usage: 612.0 bytes

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	YearsExperience	Salary
count	30.00000	30.000000
mean	5.57000	83113.333333
std	2.92211	28988.734237
min	1.10000	39200.000000
25%	3.32500	59925.000000
50%	5.10000	79350.000000
75%	8.12500	106275.000000
max	10.50000	135000.000000

0.9965480200876051

0.9986651018711702

[[9917.57201802]]

[27760.44923723]

Enter Years of Experience: 5

Estimated Salary for 5.0 years of experience is [[77348.30932733]]:

